Example 2: Fetch data from Statistics Estonia with R

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Setup R

```r
library(quarto)     # for compiling Quarto presentations and documents
library(pxweb) # query and fetch data from Statistics Estonia
library(tidyverse)  # dplyr, ggplot2, readr, etc.
library(here)  # path settings
```

Accessing Statistics Estonia Data with `pxweb`

- Statistics Estonia is **not yet part of DBnomics**
- There is **no dedicated R package** for andmed.stat.ee
- **However**, you can access the data using the `pxweb` R package
- It is based on the **PX-WEB API**, also used by many Nordic statistical agencies

Data Access Options

- **Interactive Access**
  Browse tables and build queries via the Statistics Estonia website in the R console
- **Direct Programmatic Query**
  Use the API endpoint (a *URL*) and supply **key-value pairs** to specify:
  - Variables (e.g., indicators, time, sector)
  - Values or selections for each variable

Find and download data interactively

```r
pxweb_interactive() # All available databases
pxweb_interactive("https://andmed.stat.ee/api/v1/en/stat") # Statistics Estonia
```

```
#| eval: false
==================================================================================
 R PXWEB: Content of 'andmed.stat.ee'
          at '/api/v1/en/stat'
==================================================================================
 [ 1 ] : Environment
 [ 2 ] : Economy
 [ 3 ] : Population
 [ 4 ] : Social life
 [ 5 ] : Multidomain statistics
 [ 6 ] : Population and Housing Census
 [ 7 ] : Discontinued datasets
==================================================================================
Enter your choice:
('esc' = Quit, 'b' = Back, 'i' = Show id)
1:
```

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GET call to PXWEB API

- Example dataset: PA111 - AVERAGE MONTHLY GROSS WAGES (SALARIES)

```r
pa111_meta <- pxweb_get(
  url = "https://andmed.stat.ee/api/v1/en/stat/PA111")
pa111_meta
```

```
PXWEB METADATA
PA111: AVERAGE MONTHLY GROSS WAGES (SALARIES), MEDIAN, DECILES AND NUMBER OF EMPLOYEES BY ECONOMIC ACTIVITY
SECTION (QUARTERLY)
variables:
 [[1]] Näitaja: Indicator
 [[2]] Tegevusala: Economic activity
 [[3]] Vaatlusperiood: Reference period
```

The object is just metadata, no data yet!

How to identify key-value pairs?

- Use either:
  1. Programmatically via the metadata object
  2. From the Statistics Estonia web interface – click “**API query for this table**”

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Key-value pairs

- Values for the first key (Näitaja)

```r
pa111_meta$variables[[1]]
```

```
$code
[1] "Näitaja"

$text
[1] "Indicator"

$values
 [1] "GR_W_AVG"    "NR_EMPL"     "GR_W_D1"     "GR_W_D2"     "GR_W_D3"
 [6] "GR_W_D4"     "GR_W_D5"     "GR_W_D6"     "GR_W_D7"     "GR_W_D8"
[11] "GR_W_D9"     "GR_W_AVG_SM"

$valueTexts
 [1] "Average monthly gross wages (salaries), euros"
 [2] "Number of employees"
 [3] "1st decile of monthly gross wages (salaries), euros"
 [4] "2nd decile of monthly gross wages (salaries), euros"
 [5] "3rd decile of monthly gross wages (salaries), euros"
 [6] "4th decile of monthly gross wages (salaries), euros"
 [7] "Median (5th decile) of monthly gross wages (salaries), euros"
 [8] "6th decile of monthly gross wages (salaries), euros"
```

Fetch data

- Set: Key-value pair (as R list object) & Url to the database
- Wildcard operator * to select all values

```r
px_query_list1 <- list(
  "Näitaja" = c("GR_W_D1", "GR_W_D5", "GR_W_D9"),
  "Tegevusala" = "*",
  "Vaatlusperiood" = "*"
)

pa111 <- pxweb_get(
  url = "https://andmed.stat.ee/api/v1/en/stat/PA111",
  query = px_query_list1
)
```

- We get a list object with the data

```r
pa111
```

```
PXWEB DATA
With 4 variables and 960 observations.
```

```r
class(pa111)
```

```
[1] "pxweb_data" "list"
```

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Convert the pxweb list object to a data frame

```r
# Convert to data frame
pa111.df <- as.data.frame(pa111, column.name.type = "text", variable.value.type = "text")
pa111.code <- as.data.frame(pa111, column.name.type = "code", variable.value.type = "code")
# Optional: Combine text and code in one data frame
pa111.df$Näitaja <- pa111.code$Näitaja
```

```r
head(pa111.df, n=3)
```

```
                                          Indicator      Economic activity
1 1st decile of monthly gross wages (salaries), euros Total - all activities
2 1st decile of monthly gross wages (salaries), euros Total - all activities
3 1st decile of monthly gross wages (salaries), euros Total - all activities
  Reference period
1          2021 Q1
2          2021 Q2
3          2021 Q3
  PA111: AVERAGE MONTHLY GROSS WAGES (SALARIES), MEDIAN, DECILES AND NUMBER OF EMPLOYEES
1                                                                                    532
2                                                                                    472
3                                                                                    508
  Näitaja
1 GR_W_D1
2 GR_W_D1
3 GR_W_D1
```

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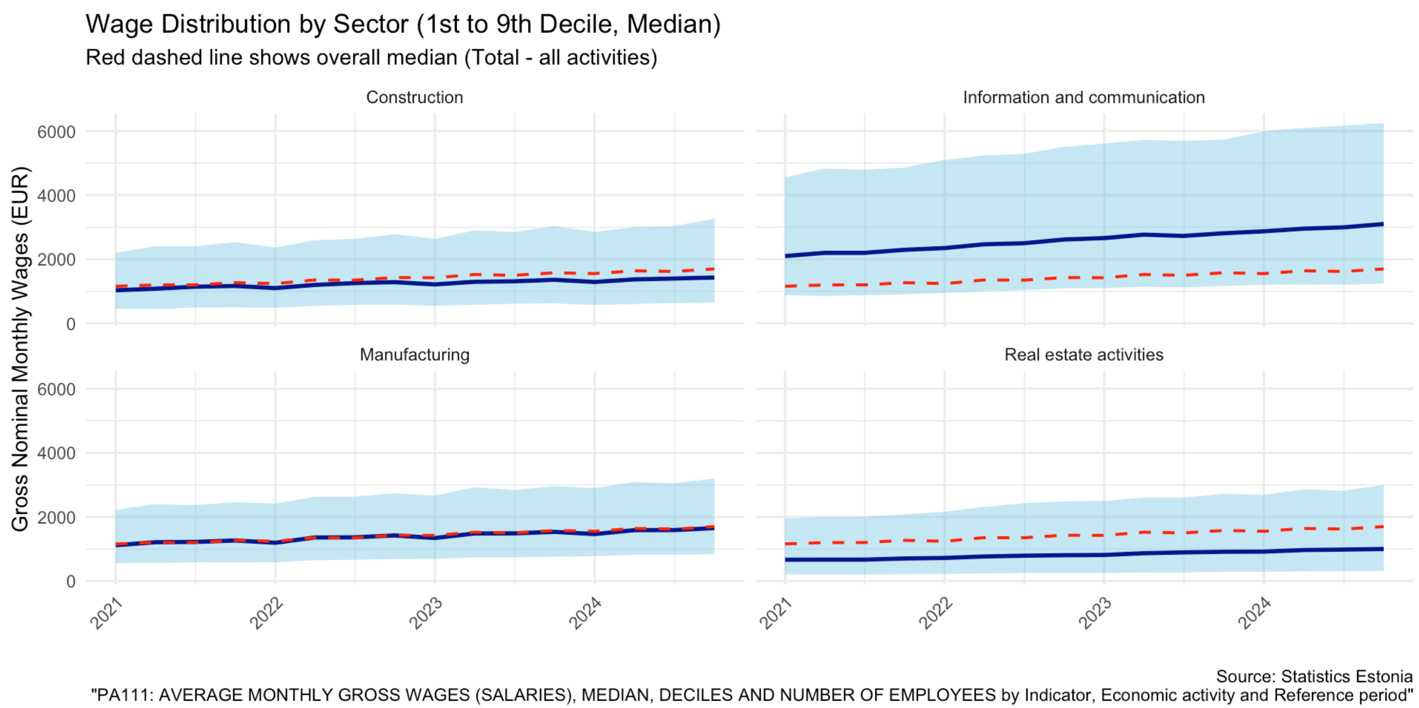
```r
ggplot(pa111_plotdata, aes(x = quarter)) +
  geom_ribbon(aes(ymin = D1, ymax = D9), fill = "skyblue", alpha = 0.5) +
  geom_line(aes(y = Median), color = "darkblue", linewidth = 1) +
  geom_line(aes(y = total_median), color = "red", linetype = "dashed", linewidth = 0.7) +
  facet_wrap(~`Economic activity`, ncol = 2) +
  labs(
    title = "Wage Distribution by Sector (1st to 9th Decile, Median)",
    subtitle = "Red dashed line shows overall median (Total - all activities)",
     caption = source_note,
    x = "",
    y = "Gross Nominal Monthly Wages (EUR)"
  ) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



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Rendering All Formats from R or Bash

To render the document to **all output formats** (docx, Beamer, Reveal.js): run programmatically from another script or the R console:

```r
quarto::quarto_render(here("teaching_materials", "example2_statistics_estonia.qmd"), output_format = "all")
```

or from the command line:

```bash
quarto render example2_statistics_estonia.qmd
```

End of session